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Robot project for Robocub / RoboCup at home 2026

sorry this is not organized, just my ideas so you understand what i need

Hi Omar,

I want to make a robot for the Robocub atHome 2026 competition. I am not technical so I do not know the exact words, but I need a robot that can enter a house / room and act smart like the videos online.

The robot should move by itself without hitting chairs or people. It should know where it is, like kitchen, living room, bedroom, maybe hallway. If someone tells it go to the kitchen it should go. If the map changes a little it should not get lost immediately.

I think it needs camera and maybe lidar, microphone, speaker, battery, and maybe a hand/arm if that is useful. But I do not know if arm is needed or too expensive. Please choose what is enough for competition.

Main idea: robot should find guests, ask them their name, remember where each guest is sitting/standing, and say something like Ahmed is near the sofa and Sara is beside the table. Also if possible know features like wearing hat, beard, glasses, red shirt, etc.

important: stable > fancy

no demo crash please

also logs for debugging?

~~maybe make it fly? no sorry ignore this~~

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More random requirements:

- Voice: it should talk in English for judges. Arabic would be nice for our team but not first priority.
- It should answer simple questions if judge asks something. I do not know if this is local AI or internet, but internet maybe not allowed? please check.
- It should recognize objects: bottle, cup, chair, table, person, maybe medicine box, bag, fruits. Not all objects, just enough to show good demo.
- It should follow a person if needed, but safely and not too close.
- It should have a laptop dashboard to see camera, battery, map, current task, and errors. Maybe a start/stop button.
- I want the robot to save videos or logs because when it fails we need to know why.

Competition things I am not sure about:

1. Is it called RoboCup@Home 2026 or Robocup at home? I wrote Robocub by mistake maybe.
2. What are the exact tasks? Please read rules and tell me what features we should build first.
3. Can we use TurtleBot 4 or should we build from zero? I heard TurtleBot is easier.
4. Do we need face recognition? If privacy problem then maybe just name + clothes.
5. Should robot use YOLO, ROS, navigation, speech recognition? I do not know, choose best stack.

Budget: not clear yet. Please make phases:

Phase 1: basic robot moves and maps

Phase 2: people detection and talking

Phase 3: competition tasks and polishing

Phase 4: optional arm / object pickup if needed

Deadline: need something working before competition, not last week. Prefer weekly progress and videos.

Please convert this messy idea into a real requirement doc and tell me missing questions only if very important. I do not want too many technical questions.

Thank you,

Client

Sent from my iPhone

NOTE

I attached videos in WhatsApp
but maybe not useful
just make same idea